Minimum Cost Flow Problem

Production/Distribution Problem

Notation and Assumptions

Let $G = (N, A)$ be a directed network with a *cost* c_{ij} and a *capacity* u_{ij} associated with every arc $(i, j) \in A$. We associate with each node $i \in N$ a number $b(i)$ which indicates its supply or demand depending on whether $b(i) > 0$ or $b(i) < 0$. The minimum cost flow problem can be stated as follows:

$$\text{Minimize} \quad z(x) = \sum_{(i,j)\in A} c_{ij}x_{ij} \tag{9.1a}$$

subject to

$$\sum_{\{j:(i,j)\in A\}} x_{ij} - \sum_{\{j:(j,i)\in A\}} x_{ji} = b(i) \qquad \text{for all } i \in N, \tag{9.1b}$$

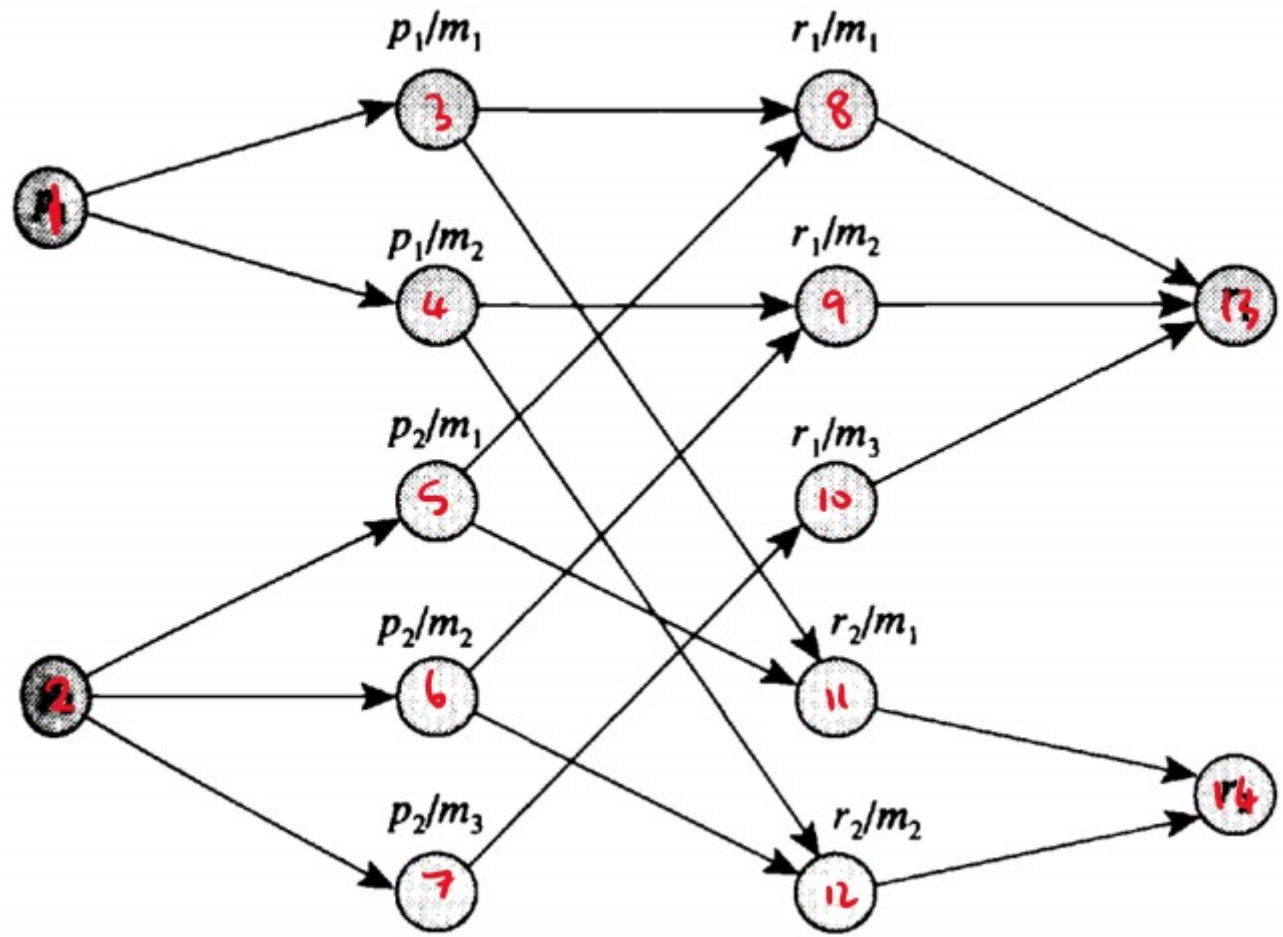
$$0 \le x_{ij} \le u_{ij} \qquad \text{for all } (i, j) \in A. \tag{9.1c}$$

$$\text{Minimize} \quad z(x) = \sum_{(i,j)\in A} c_{ij}x_{ij}$$

subject to

$$\sum_{\{j:(i,j)\in A\}} x_{ij} - \sum_{\{j:(j,i)\in A\}} x_{ji} = b(i) \qquad \text{for all } i \in N,$$

$$0 \le x_{ij} \le u_{ij} \qquad \text{for all } (i, j) \in A.$$



Production/Distribution Problem

- A car manufacturer has several manufacturing plants and produces several car models at each plant that it then ships to geographically dispersed retail centers throughout the country.
- Each retail center requests a specific number of cars of each model.
- The firm must determine the production plan of each model at each plant and a shipping pattern that satisfies the demands of each retail center and minimizes the overall cost of production and transportation.

Production/Distribution Demands and Costs

Supply/Demand

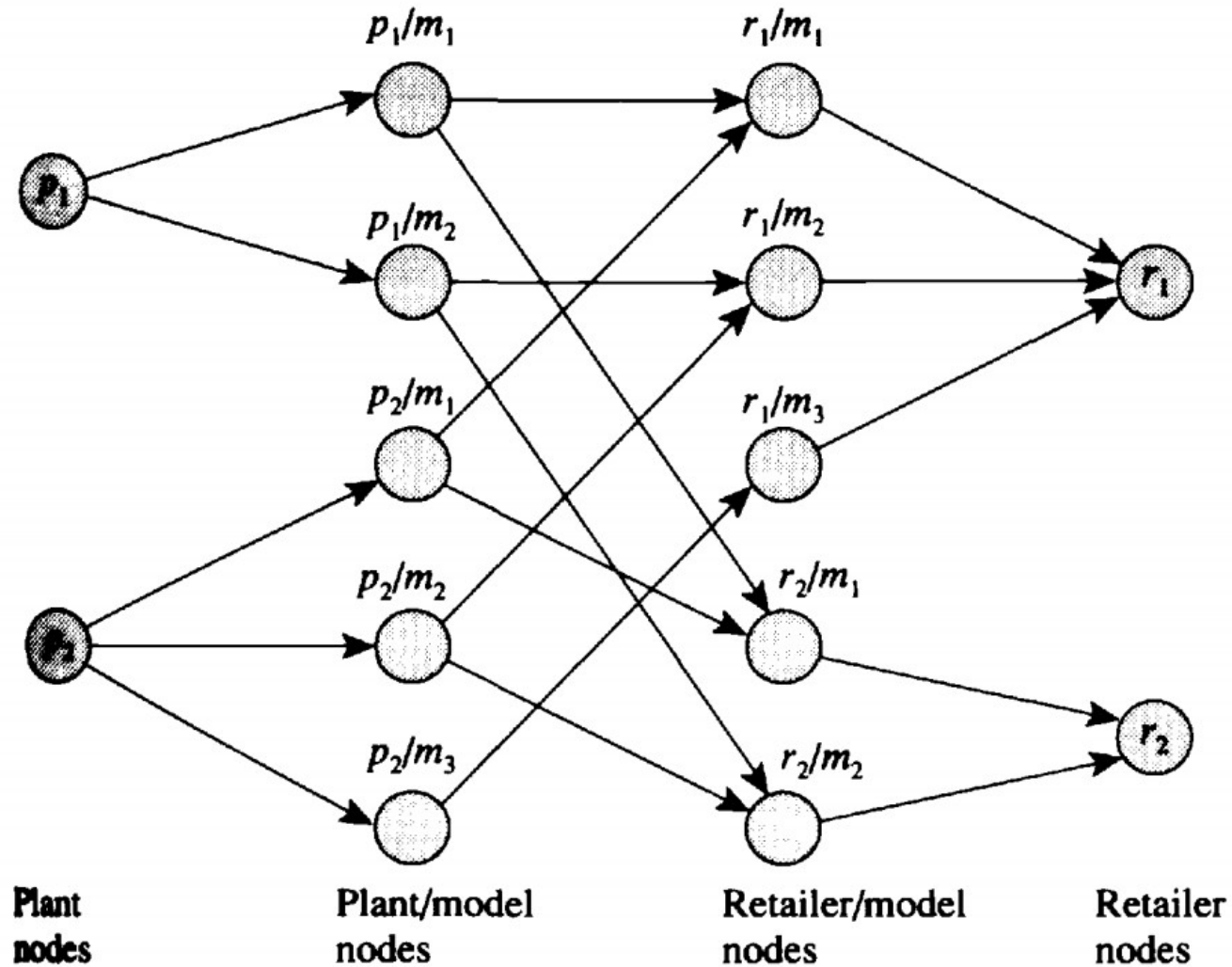
	Plant 1	Plant 2	Retailer 1	Retailer 2
Model 1	150	100	100	100
Model 2	120	150	80	130
Model 3	-	100	70	-

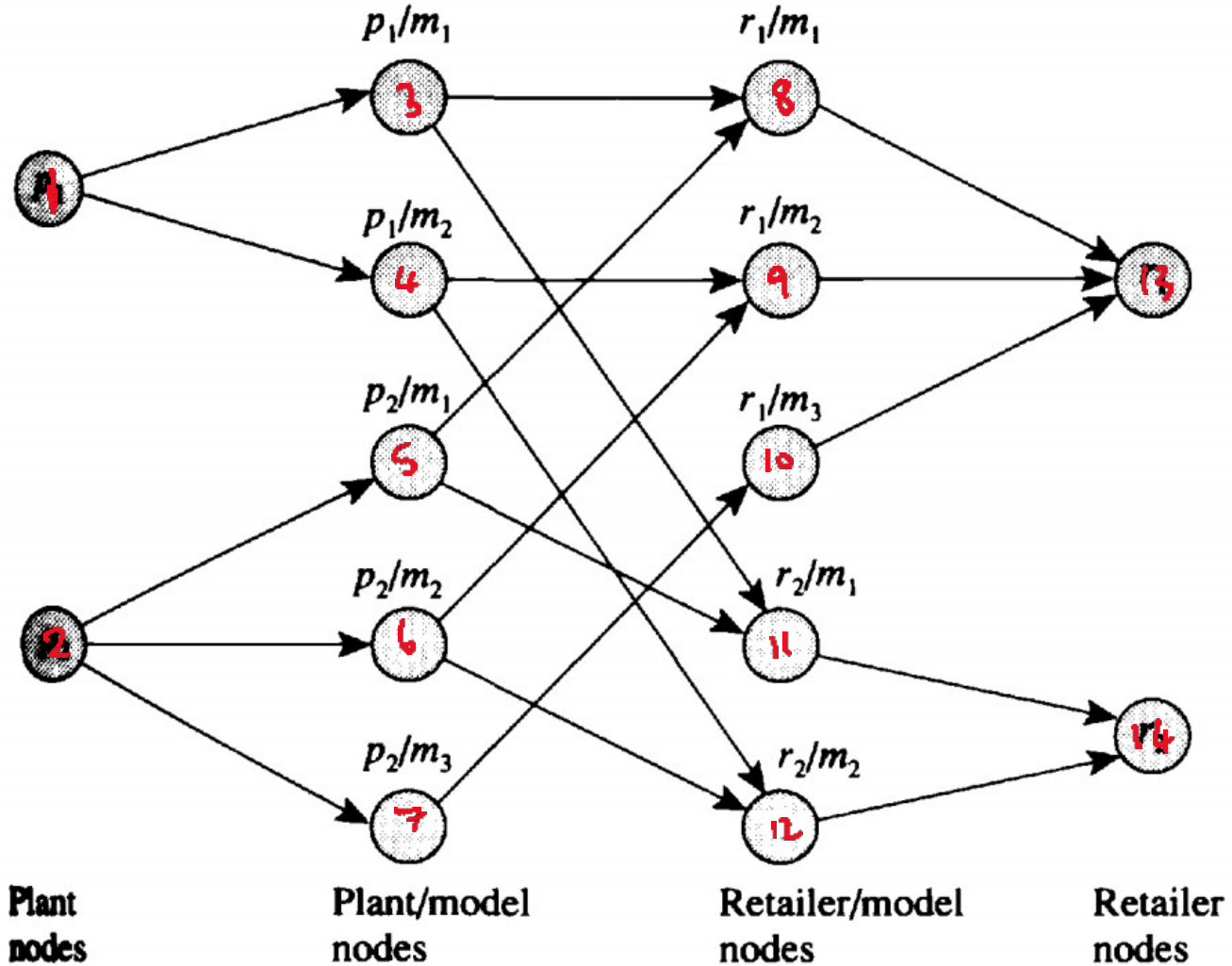
Cost of Production

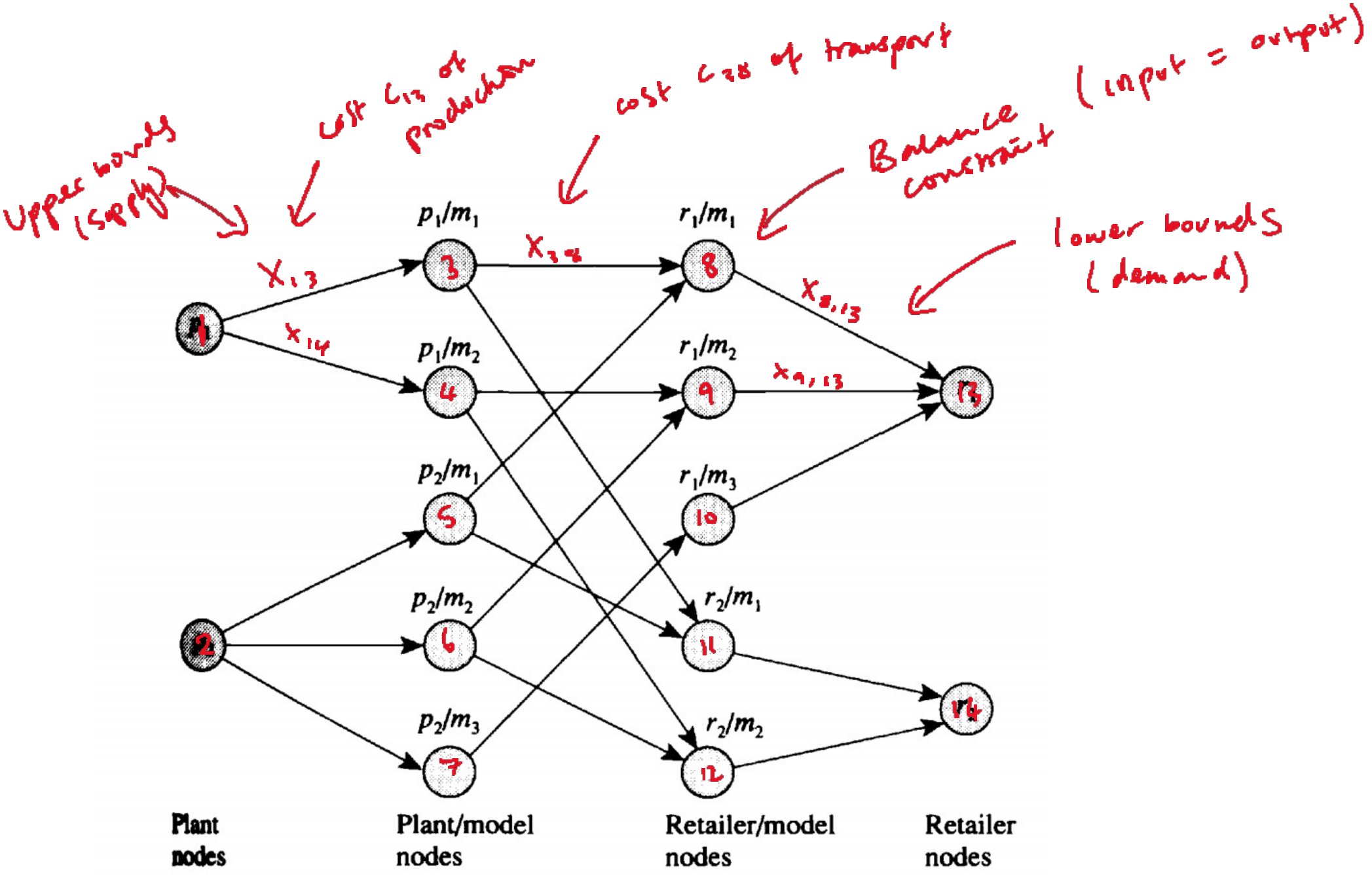
	Plant 1	Plant 2
Model 1	1000	900
Model 2	1200	1100
Model 3	-	1500

Cost of Transportation

	Plant 1			Plant 2		
	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3
Retail 1	4	3	-	5	4	4
Retail 2	5	6	-	6	3	-


p_1/m_1
r_1/m_1
p_1
p_1/m_2
r_1/m_2
r_1
p_2/m_1
r_1/m_3
p_2/m_2
r_2/m_1
r_2
p_2/m_3
r_2/m_2
Plant nodes
Plant/model nodes
Retailer/model nodes
Retailer nodes


p_1/m_1
r_1/m_1
p_1/m_2
r_1/m_2
p_2/m_1
r_1/m_3
p_2/m_2
r_2/m_1
p_2/m_3
r_2/m_2
1
2
3
4
5
6
7
8
9
10
11
12
13
14
Plant nodes
Plant/model nodes
Retailer/model nodes
Retailer nodes


Upper bounds (supply)
Unit c_{ij} of production
cost c_{ab} of transport
Balance constraint (input = output)
lower bounds (demand)
p_1/m_1
p_1/m_2
p_2/m_1
p_2/m_2
p_2/m_3
r_1/m_1
r_1/m_2
r_1/m_3
r_2/m_1
r_2/m_2
x_{13}
x_{14}
x_{38}
$x_{8,13}$
$x_{9,13}$
1
2
3
4
5
6
7
8
9
10
11
12
13
14
Plant nodes
Plant/model nodes
Retailer/model nodes
Retailer nodes

Production/Distribution Demands and Costs

upper lower bounds

	Plant 1	Plant 2	Retailer 1	Retailer 2
Model 1	150	100	100	100
Model 2	120	150	80	130
Model 3	-	100	70	-

Production costs

	Plant 1	Plant 2
Model 1	1000	900
Model 2	1200	1100
Model 3	-	1500

Transport costs

	Plant 1			Plant 2		
	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3
Retail 1	4	3	-	5	4	4
Retail 2	5	6	-	6	3	-

minimize :

```
   1000*x(1, 3) + 1200*x(1, 4) + 900*x(2, 5) + 1100*x(2, 6) + 1500*x(2, 7)
 + 4*x(3, 8) + 5*x(5, 8) + 3*x(4, 9) + 4*x(6, 9) + 4*x(7, 10) + 5*x(3, 11)
 + 6*x(5, 11) + 6*x(4, 12) + 3*x(6, 12)
```

subject to balance:

```
   -x(1, 3) + x(3, 8) + x(3, 11) == 0
   -x(1, 4) + x(4, 9) + x(4, 12) == 0
   -x(2, 5) + x(5, 8) + x(5, 11) == 0
   -x(2, 6) + x(6, 9) + x(6, 12) == 0
   -x(2, 7) + x(7, 10) == 0
   -x(3, 8) - x(5, 8) + x(8, 13) == 0
   -x(4, 9) - x(6, 9) + x(9, 13) == 0
   -x(7, 10) + x(10, 13) == 0
   -x(3, 11) - x(5, 11) + x(11, 14) == 0
   -x(4, 12) - x(6, 12) + x(12, 14) == 0
```

$x_{ij} \geq 0 \quad \forall (i,j)$

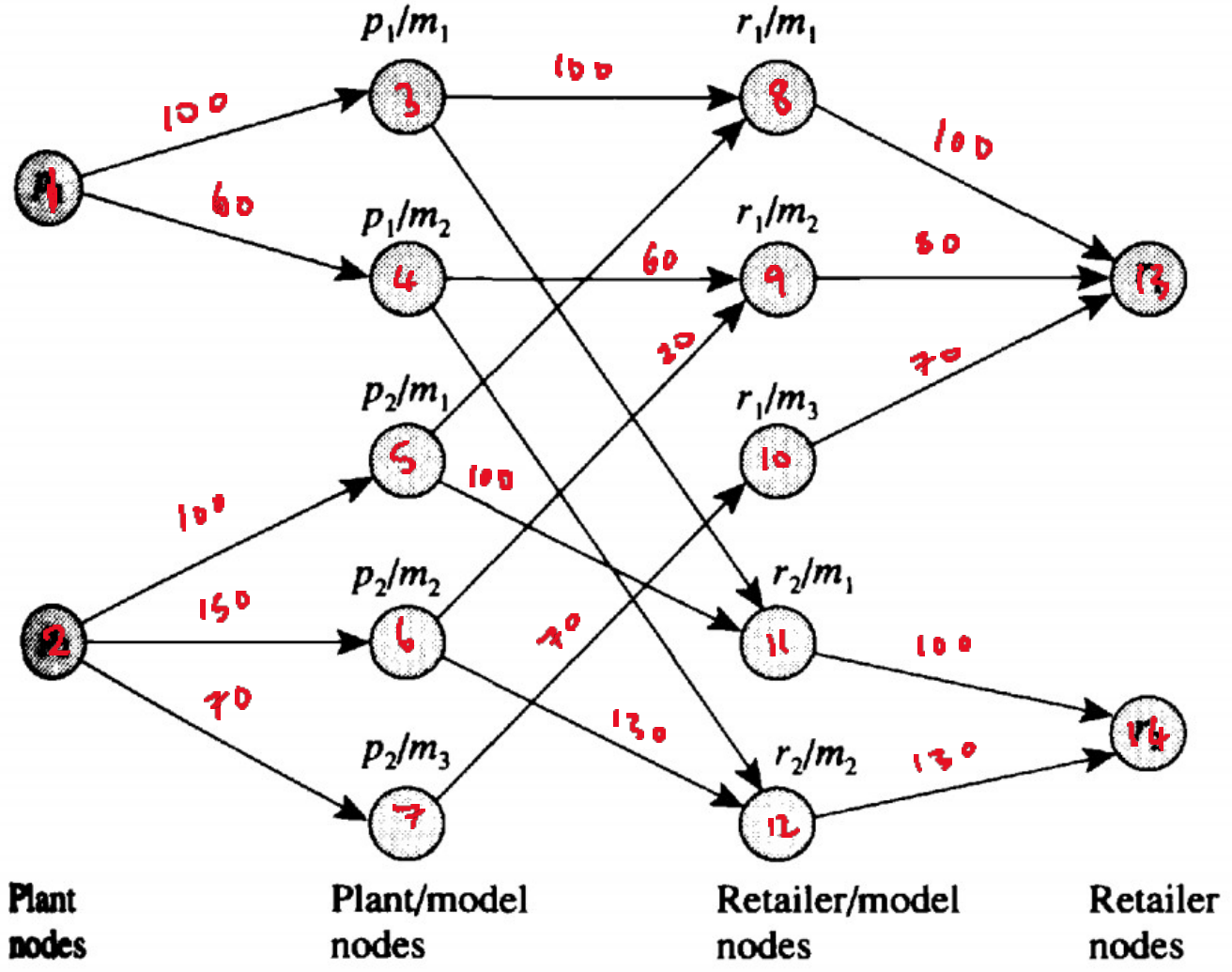
subject to supply:

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   x(1, 3) <= 150
   x(1, 4) <= 120
   x(2, 5) <= 100
   x(2, 6) <= 150
   x(2, 7) <= 100
```

subject to demand:

```
   x(8, 13) >= 100
   x(9, 13) >= 80
   x(10, 13) >= 70
   x(11, 14) >= 100
   x(12, 14) >= 130
```

Optimal Solution Via MATLAB



Cost

£523,930